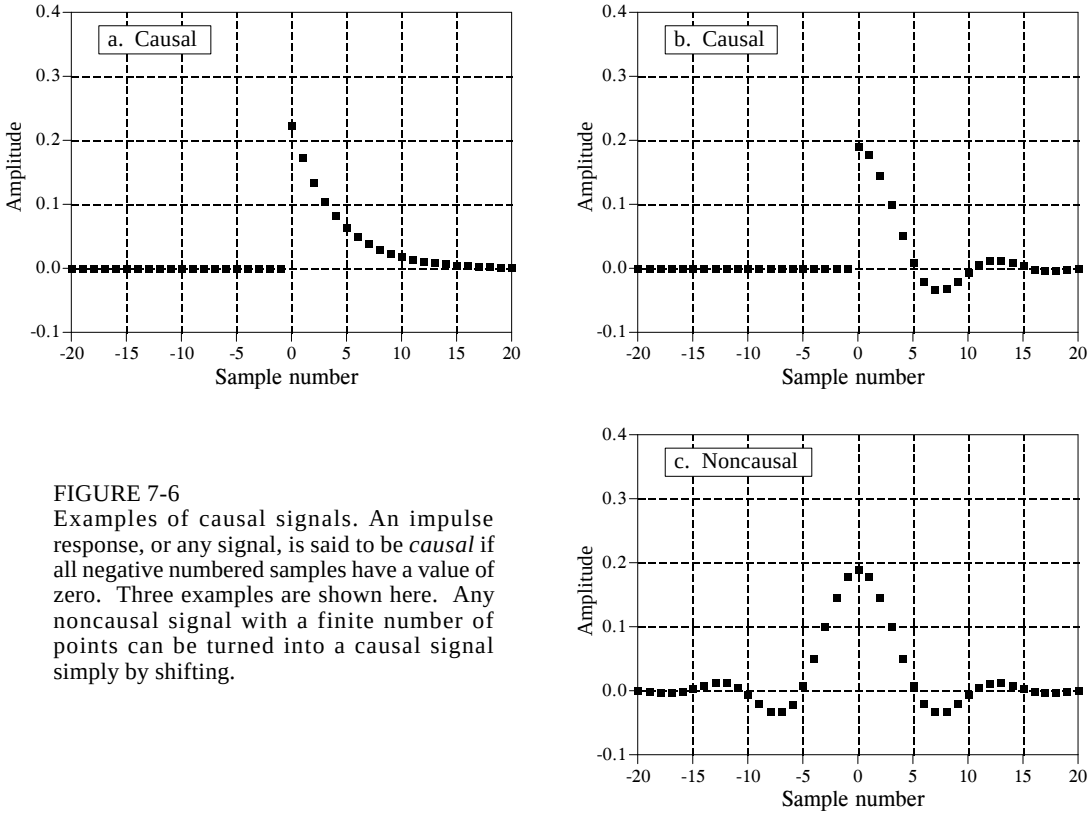

FIGURE 7-6
Examples of causal signals. An impulse response, or any signal, is said to be *causal* if all negative numbered samples have a value of zero. Three examples are shown here. Any noncausal signal with a finite number of points can be turned into a causal signal simply by shifting.

Causal and Noncausal Signals

Imagine a simple analog electronic circuit. If you apply a short pulse to the input, you will see a response on the output. This is the kind of cause and effect that our universe is based on. One thing we definitely know: *any effect must happen after the cause*. This is a basic characteristic of what we call *time*. Now compare this to a DSP system that changes an input signal into an output signal, both stored in arrays in a computer. If this mimics a real world system, it must follow the same principle of *causality* as the real world does. For example, the value at sample number eight in the input signal can only affect sample number eight or greater in the output signal. Systems that operate in this manner are said to be **causal.** Of course, digital processing doesn't necessarily have to function this way. Since both the input and output signals are arrays of numbers stored in a computer, any of the input signal values can affect any of the output signal values.

As shown by the examples in Fig. 7-6, the impulse response of a causal system must have a value of zero for all *negative numbered* samples. Think of this from the input side view of convolution. To be causal, an impulse in the input signal at sample number *n* must only affect those points in the output signal with a sample number of *n* or greater. In common usage, the term *causal* is applied to *any* signal where all the negative numbered samples have a value of zero, whether it is an impulse response or not.